1.7 Basic Matrix Operations

Name _____ Period ____ Date _____

Algebra II 1.7 (30-minute plan) · Student sheet

I can…

- add and subtract matrices.
- multiply a matrix by a scalar and evaluate matrix expressions.
- solve matrix equations.

Vocabulary · fill in the missing word

Equal matrices — same dimensions and equal _____ entries

Matrix addition and subtraction — add or subtract _____ entries; the matrices must have the same dimensions

Scalar multiplication — multiply every _____ by the number (the scalar)

Zero matrix — every _____ is 0

Part 1 · Notes and examples (fill in each note, then work the example)

Adding matrices: Matrices can be added only when they have the same _____.

Adding matrices: To add, combine _____ entries.

Example 1

A = [3 −1; 0 4] and B = [−2 5; 6 −3]. Find A + B and A − B.

Scalar multiplication: Scalar multiplication: multiply _____ entry by the scalar.

Example 2

Find −3 · [2 0; −1 4].

Example 3

Solve for x, y, and z: [x 4; −2 y] + [3 1; z 2] = [10 5; 0 −3]

Part 2 · Partner practice

Partner practice

Find [4 −2; 1 0] − [−1 3; 5 2].